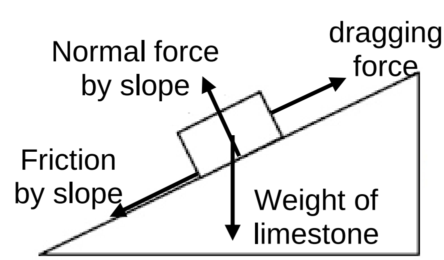
8	(a)	Work = Increase in the GPE in raising the mass to ¼ of the height $= mg \times \frac{1}{4} h = \rho \times \frac{1}{3} b^2 h \times g \times \frac{1}{4} h$ $= 2600 \times \frac{1}{3} \times 230^2 \times 147 \times 9.81 \times \frac{1}{4} \times 147$ $= 2.4 \times 10^{12}$ J	C1 A1
		Marker's comment Poorly done. Many are not aware that the work done in lifting the limestones is equivalent to the increase in GPE needed. For those who did, many did not understand that the correct height is a quarter of its height of the pyramid, information from the passage. Some also failed to use the volume to determine the mass of the limestones involved.	
	(b)	**(i)** <u>Negative work is done by friction</u> when dragging the stone up the slope.	B1
		Marker's comment Average. Students needed to be clear in their explanation to explain the effect that friction has on the work, instead of merely stating the presence of friction.	
		(ii)  All correct and none incorrect 2/2 2 correct and 2 incorrect **<u>or</u>** only 2 correct 1/2	B2
		Marker's comment Average. Several did not understand that a dragging force is needed to drag the limestone up the ramp while others did not follow the instruction of labelling the forces involved.	
		(iii) Since the stone is moving up the ramp at constant speed, $N = mg \cos \theta$ $\therefore f = \mu N = \mu\, mg \cos \theta$ $f = 0.75 \times 2500 \times 9.81 \cos 5°$ $= 1.83 \times 10^4$ N $\approx 1.8 \times 10^4$ N	M1 A1
		Marker's comment Below satisfactory. Many did not use the free-body diagram to find out the relationship between normal force and weight. They wrongly suggest that the two values are the same or they are unable to resolve the component of weight perpendicular to the ramp surface (e.g. W sin θ). Some wrongly use 20° instead of the 5.0° which is stated in question stem of (b).	

		(iv)	$F_{\text{dragging}} = f + W_{//}$ $= \mu\, mg \cos\theta + mg \sin\theta$ $= 1.83 \times 10^4 + 2500 \times 9.81 \sin 5.0°$ $= 2.04 \times 10^4\ \text{N} \approx 2.0 \times 10^4\ \text{N}$	M1 A0
		Marker's comment Below satisfactory. Since many are unable to find the correct friction in (b)(iii), they fail to show how the friction and component of weight along ramp will add up to equate to the dragging force. Again, some are unable to resolve the component of the weight parallel to the ramp surface.		
		(v)	Pulleys can be placed on the posts so that more people can be pulling downwards.	B1
		Marker's comment Poorly done. A wide variety of answers is seen, with many suggesting that the posts are meant to stabilise the ramp from moving or stop the blocks while the people are resting. Students fail to address <u>how it is easier</u> to drag the blocks up.		
	(c)	(i)	$W = N \times F_{dragging} \times s$ $= 2.7 \times 10^6 \times 2.04 \times 10^4 \times \dfrac{147}{4 \sin 5.0°}$ $= 2.33 \times 10^{13}\ \text{J} \approx 2.3 \times 10^{13}\ \text{J}$	C1 A1
		Marker's comment Below satisfactory. Many did not know or calculate the correct displacement undertaken when the blocks are dragged up the ramp, instead they wrongly use the quarter of the pyramid height as the displacement. Again, there are some who are unable to use trigo to find the correct displacement. Additionally, some only find the work done to drag one block instead of all the blockes.		
		(ii)	Work = number of people × work done by 1 man in 20 years $2.33 \times 10^{13} = N \times 0.20 \times 870 \times 4.18 \times 10^3 \times 365 \times 20$ $N = 4390 \approx 4400$	C1 A1
		Marker's comment Poorly done. The answers found in (c)(i) has made it challenging for students to do the next part. Many of the answers found in (c)(i) is too small, making the students doubt their small answer in (c)(ii). Many also have difficulty piecing up all the necessary data to compute. A significant majority of the students only use the energy available by one man in 1 year and fail to recognise that the building of pyramid took 20 years.		